RESEARCH ON FACTORS INFLUENCING TRUST IN BIOMETRIC AUTHENTICATION AND CONTINUANCE INTENTION TO USE THE MBBANK APP AMONG INDIVIDUAL CUSTOMERS IN HO CHI MINH CITY

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Từ khóa:
MBBank; xác thực sinh trắc học; niềm tin; nỗ lực xác thực; năng lực tự thực hiện; ý định tiếp tục; PLS-SEM.

TÓM TẮT

Nghiên cứu xem xét vì sao một lớp xác thực được thiết kế để bảo vệ tài khoản có thể đồng thời trở thành điểm ma sát trong trải nghiệm ngân hàng số. Nghiên cứu phân tích 1.187 hồ sơ theo cấu trúc bảng hỏi được phân tích bằng Cronbach's Alpha, EFA, PLS-SEM cùng các kiểm tra dự báo và độ bền. Kết quả cho thấy khách hàng không hình thành niềm tin từ một tín hiệu duy nhất: họ đồng thời đánh giá khả năng bảo vệ, độ ổn định khi xác thực, mức rõ ràng của quy trình và cảm giác còn kiểm soát được dữ liệu sinh trắc học. Niềm tin giúp duy trì ý định sử dụng, nhưng nỗ lực thao tác tạo một chi phí riêng mà niềm tin không tự xóa bỏ. Điểm quan trọng hơn là cùng một quy trình không gây tổn hại như nhau cho mọi người: năng lực tự thực hiện cao làm tác động bất lợi của nỗ lực yếu đi rõ rệt. Vì vậy, bài toán của MB không phải chọn giữa an toàn và thuận tiện, mà là giữ lớp bảo vệ cần thiết trong khi loại bỏ vòng lặp, giải thích đúng bước kế tiếp và hỗ trợ mạnh hơn cho người dùng khó tự khắc phục lỗi.

Keywords:
MBBank; biometric authentication; trust; authentication effort; biometric self-efficacy; continuance intention; PLS-SEM.

ABSTRACT

This study examines why an authentication layer designed to protect user accounts can simultaneously become a source of friction in the digital banking experience. The study analyzes 1,187 survey responses using Cronbach's Alpha, exploratory factor analysis (EFA), PLS-SEM, as well as predictive and robustness assessments. The findings show that customers do not develop trust from a single signal; instead, they simultaneously evaluate the level of protection provided, the reliability of the authentication process, the clarity of procedural guidance, and the extent to which they feel they retain control over their biometric data. Trust helps sustain continued usage intention, but the effort required to complete authentication creates a distinct cost that trust alone cannot eliminate. More importantly, the same process does not affect all users equally: stronger self-service capability substantially weakens the adverse effect of required effort. Therefore, MB's challenge is not to choose between security and convenience, but to preserve the necessary level of protection while eliminating repetitive steps, clearly explaining what users should do next, and providing stronger support for those who have difficulty resolving authentication problems on their own.

1. INTRODUCTION

July 01, 2024 marked a very specific change in the digital banking experience in Vietnam: for many online transactions, customers had to prove not only that they knew the password or OTP, but also that the person holding the device was the account holder. Decision 2345/QĐ-NHNN introduced biometric authentication into high-value transaction situations and device changes, turning facial and identity-card data from a supporting feature into a control layer positioned immediately before the money-transfer action (State Bank of Vietnam, 2023).

For MB, this touchpoint is large enough in scale for a small error to become a significant operating cost. At the end of 2024, MB announced approximately 30 million customers, with most transactions taking place on digital channels, and at times the MBBank App processed tens of millions of transactions per day (MB, 2024). When an authentication step is repeated at that scale, a few rescans are not merely a few seconds lost. They accumulate into waiting time, delayed transactions, support requests, and the possibility that customers shift everyday needs to another application.

Therefore, the problem does not lie in choosing "security" or "ease of use" as two mutually exclusive goals. An authentication layer can prevent fraud yet still weaken the experience if recognition is unstable, instructions are ambiguous, or customers do not know how far sensitive data are used. Conversely, a very fast process that does not create a sense of protection is also insufficient for users to rely on when transactions involve money and identity. The practical management question is how to maintain the required level of protection without forcing customers to pay an additional invisible cost in time, effort, and a sense of loss of control.

Studies in Vietnam have addressed both mobile-banking continuance intention and biometric authentication, but each study has resolved only part of the decision chain. Nguyen and Dao (2024), with 523 customers of major banks in Vietnam, showed that perceived usefulness, satisfaction, adaptation, and self-efficacy were significantly associated with continuance intention; trust also changed the adaptation → continuance intention relationship. More specifically, Minh Phuong Gia Hoang and Shon Hoang (2024) surveyed 313 Gen Z respondents in Ho Chi Minh City and found that consumer trust was associated with continuance usage intention for biometric authentication, while also placing perceived risk in a mediating mechanism. These two research directions bring the question closer to the post-adoption authentication experience, but they have not separated within the same model the signals that form trust, the distinct cost of effort, and the ability of self-efficacy to change the extent to which effort accompanies continuance intention.

The gap lies in the post-adoption stage, when customers have already had to use biometrics in a compliance-oriented banking process. The studies found have not directly answered three successive questions: which signals customers rely on to decide whether an authentication step is trustworthy; whether operational effort creates a separate cost path to continuance intention; and whether the same level of effort causes equal harm to users with different levels of self-handling capability.

From that gap, the study "Factors influencing Trust in biometric authentication and Continuance intention to use the MBBank App among individual customers in Ho Chi Minh City" was conducted to identify factors associated with customers' trust in the biometric authentication step, assess the relationship between trust and continuance intention to use the MBBank App, while also examining the separate relationship of operational effort with continuance intention and whether users' self-handling capability changes the strength of this relationship. The study also examines the stability of the main relationships when usage characteristics and authentication experience are taken into account, thereby providing a basis for MB to identify points that should be prioritized for improvement without weakening security requirements.

The objective of the study is to determine the degree of association of PSE, BAR, PT, and BPC with TBA; test the roles of TBA and AE with respect to CI; determine whether BSE weakens the adverse effect of AE on CI; and simultaneously assess the stability of the main relationships when age group, chip-based identity-card reading capability, number of re-authentication attempts, usage frequency, and transaction value are taken into account. The results are used to identify where MB should prioritize reducing loops, increasing stability, clarifying data rights, and guiding the next step while keeping the required security guardrails unchanged.

2. RESEARCH METHODOLOGY

2.1 Theoretical basis and research model

Bhattacherjee (2001) shifted the focus of information-systems research from the initial adoption decision to the continuance decision after users have gained experience. The post-adoption perspective is appropriate for the MBBank App because biometrics is no longer a technology that customers merely consider whether to "try or not"; it appears in specific transactions and is evaluated through what users actually experience. Tam, Santos, and Oliveira (2020) further showed that intention to continue using an application depends on the evaluation of post-use benefits and costs, so a continuance model needs to separate the value received from the effort that must be expended.

In biometric authentication, customers do not directly observe anti-spoofing algorithms, matching methods, or the entire data-processing flow. They have to infer whether the system can be relied upon from visible signals. Perceived Security Effectiveness (PSE) answers “does this step really protect me”; Biometric Authentication Reliability (BAR) answers “does the system work correctly when I need it”; Process Transparency (PT) answers “why do I have to do this step and, if it fails, what should I do next”; Biometric Privacy Concern (BPC) reflects the cost component when users feel that facial or identity-card data may move beyond their scope of control. The four signals are placed before Trust in Biometric Authentication (TBA) instead of being combined into an overly broad concept of “security.”

Perceived Authentication Effort (AE) is separated from TBA because a customer may still trust that the system is secure but not want to pay the operational cost repeatedly. Biometric Self-Efficacy (BSE) is positioned as a moderator because the same error does not create the same level of disruption for someone who knows how to adjust, retry, and follow instructions compared with someone who does not know which step needs to be corrected. Moriuchi (2021) showed that self-efficacy changes users’ responses to benefits and risks in facial-recognition payment; in Vietnam, Nguyen and Dao (2024) also found that self-efficacy was significantly associated with mobile-banking continuance intention. This evidence supports testing BSE as a condition that changes the strength of the AE → CI relationship, rather than viewing BSE merely as a background characteristic.

The model retains seven direct and moderating hypotheses: H1, PSE → TBA (+); H2, BAR → TBA (+); H3, PT → TBA (+); H4, BPC → TBA (−); H5, AE → CI (−); H6, TBA → CI (+); H7, AE × BSE → CI (+), meaning that high BSE is expected to make the negative AE → CI relationship weaker. To estimate the interaction correctly, the CI equation still retains the component effect BSE → CI but does not attach a separate hypothesis to it; H7 tests only the change in slope due to the interaction. Age group, chip-based identity-card reading capability, number of required re-authentication attempts, frequency, and transaction value are used for robustness checks and are not turned into main hypotheses.

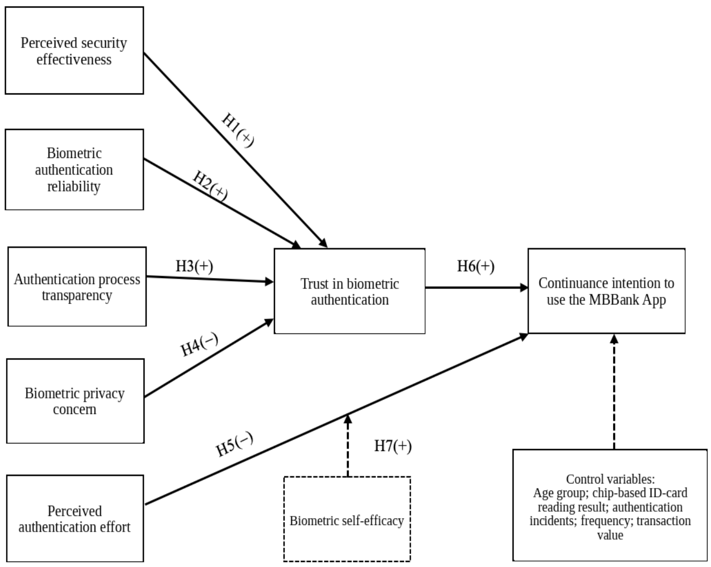

Figure 1: Proposed research model

2.2 Analysis method

The analysis is implemented in sequence from scale quality to the structural model. Step 1 uses Cronbach’s Alpha, corrected item-total correlation (CITC), and Alpha if item deleted to test internal consistency. For reflective scales, an Alpha of approximately 0,70-0,95 is regarded as an appropriate range; CITC of 0,30 or above is used as a screening threshold, but an item is removed only when statistical evidence is accompanied by a content-based reason rather than deleting a variable merely to increase Alpha (Hair et al., 2022).

Step 2 uses EFA with Principal Component Analysis and Varimax rotation for 24 variables belonging to PSE, BAR, PT, BPC, AE, and BSE. KMO greater than 0,50 together with Bartlett Sig. < 0,05 confirms that the correlation matrix is adequate; absolute factor loadings from 0,50 are prioritized in interpretation, while cross-loadings, extracted variance, and theoretical structure are considered together. TBA and CI are not forced into a separate EFA merely to create another table; the two pre-specified constructs are assessed at the measurement-model layer using loading, CR, AVE, and HTMT.

Step 3 evaluates the measurement and structural models using PLS-SEM. Loadings of approximately 0,708 or above, CR and Alpha in the 0,70-0,95 range, AVE from 0,50, and HTMT below 0,85 are used as guiding benchmarks. The structural model reports β, Bootstrap SE, 95% confidence intervals, f^2, R^2, and inner VIF; out-of-sample predictive ability is evaluated separately using PLSpredict with Q^2_predict and RMSE. CFI, TLI, RMSEA, or GFI are not included in the main tables because these indices belong to a different evaluation logic and can blur the function of PLS-SEM (Hair et al., 2022; Henseler, Ringle, and Sarstedt, 2015; Shmueli et al., 2016).

Step 4 tests moderation using the AE × BSE interaction term after standardizing the two component variables. The equation simultaneously retains AE, BSE, and AE × BSE; the β of the interaction only indicates how the AE → CI slope changes with BSE, so the conclusion also relies on simple slopes at low, medium, and high BSE together with the additional R^2 when the interaction is added to the model. Step 5 uses a One-Sample T-Test with a benchmark of 4 as the managerial expectation standard. The two adverse constructs BPC and AE are reverse-coded into BPC* = 6 − BPC and AE* = 6 − AE before comparison with this benchmark so that all higher scores carry the same favorable meaning; group differences are tested using ANOVA or t-test where appropriate. PLSpredict and clustering serve only predictive or supplementary exploratory roles and are not used as causal evidence.

2.3 Data collection method

The questionnaire structure defines the target respondents as individual customers aged 18 or older, living or working in Ho Chi Minh City, having an MB account, having used the MBBank App, and having completed biometric authentication at least three times in the most recent six months. The repeated-experience condition is imposed so that measurement does not depend on a single initial registration. A 5-point Likert scale is used for 32 observed variables belonging to eight constructs; background variables record age, length of use, frequency, transaction value, chip-based identity-card reading capability, and number of required re-authentication attempts.

The analytical dataset consists of 1.187 observations following the questionnaire structure and is used consistently for scale testing, EFA, PLS-SEM, moderation testing, group differences, and robustness checks. The minimum threshold of 1.145 was determined in advance by power analysis for a specification with eight explanatory variables, with f^2=0,02, α=0,05, and target power of 0,95; n=1.187 gives power of approximately 0,958. Determining the size according to the most demanding specification avoids dependence on simple sample-size rules that may produce imprecise estimates in PLS-SEM (Kock and Hadaya, 2018).

3. RESULTS AND DISCUSSION

3.1 Characteristics of the analytical sample

The analytical sample consists of 1.187 observations, distributed by age, length of use, frequency, transaction value, and authentication status. In the dataset, 59,8% of observations are in the status where the phone can read the chip-based identity card, while 72,1% record having had to re-authenticate at least once. These two statuses have more operational value than a purely demographic label because they lie directly within the authentication journey: one session may be completed on the first attempt, while another session must repeat the same objective before the transaction can continue.

Table 1: Characteristics of the analytical sample according to the questionnaire structure

Characteristic	Sample n=1.187		
	Frequency	Percentage %	% Cumulative
Gender			
Male	558	47,0	47,0
Female	629	53,0	100,0
Age group			
18–25 years old	263	22,2	22,2
26–35 years old	420	35,4	57,5
36–45 years old	312	26,3	83,8
46–55 years old	140	11,8	95,6
From 56 years old	52	4,4	100,0
Length of MBBank App use			
Under 6 months	108	9,1	9,1
6–under 12 months	298	25,1	34,2
1–under 3 years	413	34,8	69,0
3–under 5 years	241	20,3	89,3
From 5 years	127	10,7	100,0
Transaction frequency			
Less than 1 time/week	194	16,3	16,3
1–2 times/week	350	29,5	45,8
3–5 times/week	382	32,2	78,0
From 6 times/week	261	22,0	100,0
Transaction value			
Under 1 million	234	19,7	19,7
1–under 10 million	574	48,4	68,1
10–under 20 million	236	19,9	88,0
From 20 million	143	12,0	100,0
Chip-based identity-card reading result			
Readable	710	59,8	59,8
Not readable	150	12,6	72,5
Not performed/do not remember	327	27,5	100,0
Number of required re-authentication attempts			
None	331	27,9	27,9
1 time	267	22,5	50,4
2–3 times	297	25,0	75,4
From 4 times	292	24,6	100,0

Source: Descriptive statistics from the analytical dataset, n = 1.187, year 2026.

This distribution also reveals a point that must be kept correct in interpretation: demographics only indicate who is present in the sample; they do not indicate the cause that makes authentication difficult. In contrast, variables such as chip reading and the number of times the process has to be repeated lie directly within the experience chain, so they are later used as signals for robustness checks and operational segmentation rather than automatically treating age or gender as causes.

3.2 Scale reliability test results

Reliability is tested at the individual-item level before proceeding to EFA. Alpha for the eight scales ranges from 0,907-0,927, while the lowest CITC is 0,780. No item increases the scale Alpha when deleted; in other words, retaining variables does not depend on an overall "good-looking" Alpha but is also confirmed by item-total correlation and Alpha if item deleted. The high Alpha range, while still below 0,95, also helps reduce concern that the four statements within the same construct are merely repeating the same sentence in different words.

Table 2: Scale reliability test results

Alpha	Variable	Item–total c orrelation	Alpha if i tem deleted
PSE (0,910)	PSE1	0,805	0,881
	PSE2	0,794	0,885
	PSE3	0,808	0,880
	PSE4	0,780	0,890
BAR (0,912)	BAR1	0,815	0,880
	BAR2	0,790	0,889
	BAR3	0,811	0,882
	BAR4	0,782	0,892
PT (0,915)	PT1	0,787	0,896
	PT2	0,822	0,883
	PT3	0,805	0,889
	PT4	0,806	0,889
BPC (0,915)	BPC1	0,813	0,887
	BPC2	0,798	0,892
	BPC3	0,810	0,888
	BPC4	0,800	0,891
AE (0,907)	AE1	0,783	0,883
	AE2	0,786	0,881
	AE3	0,805	0,875
	AE4	0,786	0,881
BSE (0,911)	BSE1	0,802	0,884
	BSE2	0,801	0,884

Alpha	Variable	Item–total c orrelation	Alpha if i tem deleted
	BSE3	0,792	0,888
	BSE4	0,799	0,885
TBA (0,927)	TBA1	0,844	0,900
	TBA2	0,826	0,907
	TBA3	0,830	0,905
	TBA4	0,820	0,909
CI (0,920)	CI1	0,826	0,892
	CI2	0,812	0,897
	CI3	0,797	0,902
	CI4	0,824	0,892

Source: Cronbach's Alpha test results from the analytical dataset, n = 1.187, year 2026.

High reliability only answers whether the items within the same scale move together; it does not prove that PSE differs from BAR, PT differs from BPC, or TBA differs from CI. Therefore, Alpha is not used as the final evidence of model quality. The next step must examine factor structure, convergent validity, and especially discriminant validity.

3.3 Factor analysis and measurement-model results

The block of 24 variables belonging to PSE, BAR, PT, BPC, AE, and BSE has KMO = 0,849 and Bartlett p < 0,001. The six-factor solution explains a cumulative 79,173% of the variance. After Varimax rotation, the primary loadings are all approximately 0,877-0,901 while cross-loadings are very small. The important result is not the high variance figure by itself, but that the six groups of items align with six different questions: protection, stable operation, transparency, privacy, effort, and self-handling capability.

Table 3: Rotated factor matrix analysis results for the six explanatory and moderating variables

Variable	F1	F2	F3	F4	F5	F6
PSE1	**0,892**	0,034	-0,035	-0,005	-0,014	0,020
PSE2	**0,884**	0,057	0,009	0,020	-0,014	0,026
PSE3	**0,895**	0,032	-0,000	0,005	0,013	-0,025
PSE4	**0,877**	0,016	-0,007	0,003	-0,033	-0,011
BAR1	0,021	**0,899**	0,011	0,006	0,016	0,025
BAR2	0,036	**0,882**	0,011	-0,013	0,034	0,015
BAR3	0,037	**0,896**	-0,001	0,034	0,015	-0,013
BAR4	0,044	**0,877**	0,033	0,002	-0,015	-0,002
PT1	0,009	0,022	**0,880**	0,023	-0,004	-0,025
PT2	-0,014	0,013	**0,901**	0,062	-0,014	-0,013
PT3	-0,000	0,007	**0,890**	0,051	0,018	-0,044
PT4	-0,026	0,011	**0,891**	0,036	-0,017	-0,022
BPC1	0,008	-0,006	0,035	**0,896**	0,006	0,045
BPC2	-0,020	0,030	0,058	**0,886**	-0,033	0,011
BPC3	0,022	0,029	0,041	**0,894**	0,001	0,025
BPC4	0,012	-0,022	0,040	**0,888**	-0,013	0,017
AE1	-0,023	-0,016	-0,021	-0,012	**0,880**	0,001
AE2	-0,020	0,027	0,007	-0,014	**0,881**	0,019
AE3	0,000	0,007	-0,004	-0,014	**0,894**	0,011
AE4	-0,004	0,035	-0,000	0,001	**0,881**	0,010
BSE1	0,020	0,011	-0,031	0,034	0,009	**0,890**
BSE2	-0,002	-0,004	-0,018	0,016	0,018	**0,890**
BSE3	0,024	0,017	-0,052	0,036	-0,011	**0,883**
BSE4	-0,030	0,000	-0,003	0,013	0,024	**0,889**
Eigen.	**3,157**	**3,169**	**3,188**	**3,189**	**3,134**	**3,164**
% Var.	**13,153**	**13,204**	**13,285**	**13,289**	**13,058**	**13,185**
KMO = 0,849; Sig. Bartlett < 0,001; Cumulative = 79,173%						

Source: Exploratory factor analysis (EFA) results from the analytical dataset, n = 1.187, year 2026.

Table 4: Composite reliability, convergent validity, and discriminant validity

Scale	Loading	Alpha	CR	AVE	Highest HTMT
PSE	0,878–0,894	0,910	0,937	0,788	0,561
BAR	0,879–0,898	0,912	0,938	0,791	0,538
PT	0,882–0,903	0,915	0,940	0,796	0,431
BPC	0,888–0,897	0,915	0,940	0,797	0,351
AE	0,881–0,893	0,907	0,935	0,782	0,476
BSE	0,885–0,892	0,911	0,938	0,790	0,149
TBA	0,899–0,915	0,927	0,948	0,821	0,674
CI	0,887–0,904	0,920	0,943	0,806	0,674

Source: Measurement-model assessment results from the analytical dataset, n = 1.187, year 2026.

At the measurement-model layer, the lowest loading is still close to 0,88; CR is around 0,94 and all AVE values exceed 0,78. The highest HTMT in the entire model is 0,674 between TBA and CI, below the 0,85 threshold. This is an important check because TBA and CI are inherently close in content: customers who trust the authentication step also tend to continue using the application. HTMT shows that the two constructs are related but have not collapsed into a single concept.

3.4 PLS-SEM structural-model test results

The structural model is read through two equations. In the TBA equation, PSE, BAR, and PT have positive signs, while BPC has a negative sign; trust therefore accompanies a sense of protection, stable operation, and an understandable process, but decreases as data-related concern increases. The four coefficients PSE → TBA = 0,493; BAR → TBA = 0,450; PT → TBA = 0,428; and BPC → TBA = -0,379 are all standardized coefficients. The sum of the absolute β values is not a criterion for diagnosing multicollinearity: in the TBA equation itself, the inner VIF values of the four predictors are only in the range 1,007-1,011. In the CI equation, TBA has a positive sign while AE has a negative sign. BSE is retained as a required component effect of the moderation model but makes a very small direct contribution; the part that needs to be tested is whether the AE × BSE interaction changes the slope of AE. This structure separates two ledgers that a single "overall experience" indicator often conceals: customers may trust the system but still not want to expend too much effort each time they use it.

Table 5: Test results for the direct and moderating hypothesis paths

H	Path	β	Boot SE	95% CI	f²	Sig.
H1	PSE → TBA	0,493	0,015	[0,462; 0,523]	1,044	<0,001
H2	BAR → TBA	0,450	0,015	[0,419; 0,481]	0,872	<0,001
H3	PT → TBA	0,428	0,016	[0,397; 0,459]	0,786	<0,001
H4	BPC → TBA	-0,379	0,015	[-0,408; -0,350]	0,617	<0,001
H5	AE → CI	-0,363	0,016	[-0,394; -0,332]	0,439	<0,001
H6	TBA → CI	0,532	0,015	[0,503; 0,561]	1,036	<0,001
H7	AE × BSE → CI	0,354	0,014	[0,327; 0,382]	0,563	<0,001

Source: PLS-SEM/Bootstrap analysis results with 5.000 samples from the analytical dataset, n = 1.187, year 2026.

PSE has the largest β in the TBA equation, but β is not a managerial priority ranking. The coefficient indicates how sensitive TBA is to changes in PSE within the model; it does not indicate how far PSE currently falls short of the desired state. The fact that $|0{,}493| + |0{,}450| + |0{,}428| + |-0{,}379| = 1{,}750$ does not conflict with $R^2 = 0{,}769$, because R^2 depends simultaneously on the coefficient vector and the correlation matrix among the predictors rather than equaling the sum of the β values. In this dataset, the pairwise correlations among PSE, BAR, PT, and BPC are only about −0,017 to 0,096; therefore, the sum of the squares of the four coefficients, approximately 0,772, is also close to $R^2 = 0{,}769$. The four explanatory sources therefore nearly complement one another rather than repeat the same information. In contrast, AE retains a direct negative association with CI even when TBA is already in the equation. A customer may therefore still trust that the authentication step is "correct in terms of security" yet reduce use of the application for transactions that can be carried out elsewhere if retries, waiting, and self-diagnosis of errors keep recurring.

3.5 Test results for the moderating role of Biometric Self-Efficacy

H7 tests the question that the direct paths have not answered: whether the same level of effort is associated with the same decline in continuance intention for everyone. The positive AE × BSE coefficient shows that the negative AE → CI slope becomes less severe as BSE increases. The additional R^2 due to the interaction reaches approximately 0,124. In the same equation, the component effect BSE → CI has $f^2=0{,}006$, very small compared with $f^2=0{,}563$ for the interaction; therefore, the role of interest for BSE lies in its ability to change the extent to which friction translates into reduced usage intention, not in a large direct effect on CI.

Table 6: Explanatory quality and slopes of the moderating effect

Indicator	Value	Benchmark/direction	Significance
R² TBA	0,769	High explanatory power	Four antecedents explain TBA
R² CI	0,780	High explanatory power	TBA, AE, BSE, interaction, and controls
BSE → CI (component effect)	β≈0,042; f²=0,006	No H assigned	Very small direct contribution
ΔR² do AE×BSE	0,124	>0	Interaction adds explanatory power
AE slope when BSE is low	-0,717	Strongly negative	Friction causes substantial harm
AE slope when BSE is medium	-0,363	Negative	Harm at the baseline level
Slope AE khi BSE cao	-0,009	Near 0	BSE absorbs most of the adverse effect

Source: Moderation-role test results from the analytical dataset, n = 1.187, year 2026.

The simple slopes show the real-world meaning of H7. When BSE is low, an increase in AE is associated with a very large decrease in CI; at medium BSE, the negative relationship remains clear; when BSE is high, the slope is nearly zero. Skilled users do not "like" a complex process more; they simply know how to retry, adjust, and follow instructions, so the same obstacle is less likely to break the transaction flow. MB therefore has two different levers: fix process friction and support users at the exact point where they get stuck. Skills support is only a shock absorber; it cannot become a reason to retain a loop that adds no security.

3.6 One-Sample T-Test results

To place the results in the same managerial direction, the eight constructs are defined so that "a higher score is favorable." BPC and AE are therefore reversed into BPC* = 6 − BPC and AE* = 6 − AE, respectively reflecting reassurance about privacy and low effort. The benchmark of 4, corresponding to "Agree" on the 5-point Likert scale, is used as the managerial expectation standard rather than the mathematical midpoint.

Table 7: Favorably oriented perception scores and gaps to the 4,0 benchmark

Variable	Mean	SD	t	Sig.	Gap to 4
PSE	3,890	0,757	-4,984	<0,001	0,110
BAR	3,565	0,801	-18,703	<0,001	0,435
PT	3,366	0,832	-26,273	<0,001	0,634
BPC*	2,699	0,831	-53,939	<0,001	1,301
AE*	2,963	0,819	-43,623	<0,001	1,037
BSE	3,650	0,785	-15,382	<0,001	0,350
TBA	3,579	0,802	-18,063	<0,001	0,421
CI	3,681	0,772	-14,223	<0,001	0,319

Source: One-Sample T-Test results from the analytical dataset, n = 1.187, year 2026. BPC and AE* are reverse-coded scales used for managerial interpretation.*

The distance to the benchmark of 4 does not match the β ranking. After correct reverse coding, BPC* reaches only 2,699, which is 1,301 points below the expected benchmark; AE* reaches 2,963, still short by 1,037 points. PSE is on the opposite side with a mean of 3,890, only about 0,110 points below 4. Therefore, PSE has a strong association with trust but now has less remaining room for improvement; BPC and AE still contain more unresolved work. PT lies between the two sides: the perception level of 3,366 remains low while the PT → TBA path is still strong, so timely information may have more value than a generic security communication campaign.

3.7 ANOVA and post-hoc results by experience characteristics

Group differences are retained only when they answer an operational question, not merely because the p-value is small. The number of required re-authentication attempts creates a clear gradient: BAR decreases from 4,197 among those who never had to repeat the process to 2,842 among those with four or more repetitions; CI simultaneously decreases from 4,215 to 2,997. In contrast, differences in BSE by age are small and do not reach the 5% significance level. The result weakens segmentation by "profile" and strengthens monitoring of incident status—the point where customers actually get stuck in the process.

Table 8: Group differences with interpretive value

Analysis	Mean by group	F/t	Sig.	Effect
Number of re-authentication attempts × BAR	4,197 → 3,737 → 3,418 → 2,842	F=255,634	<0,001	η^2=0,393
Number of re-authentication attempts × TBA	3,890 → 3,732 → 3,507 → 3,161	F=53,348	<0,001	η^2=0,119
Number of re-authentication attempts × CI	4,215 → 3,929 → 3,536 → 2,997	F=220,683	<0,001	η^2=0,359
Chip readable (n=710) versus not/not yet readable (n=477) × BAR	3,717 versus 3,339	t=8,033	<0,001	d=0,484
Chip readable (n=710) versus not/not yet readable (n=477) × CI	3,825 versus 3,468	t=7,929	<0,001	d=0,475
Age group × BSE	small difference	F=2,135	0,074	η^2=0,007

Source: ANOVA and Independent Samples T-Test results from the analytical dataset, n = 1.187, year 2026.

The group whose phones can read the chip-based identity card has higher BAR and CI than the combined group consisting of those whose phones cannot read it or who have not performed the process/do not remember; this is still only a group association because device status was not randomly assigned. The managerial value lies in the operational signal: if low first-pass success is concentrated in one device group, the system-wide success rate may still look good while a portion of customers pays a very high retry cost. Age does not indicate which step a customer is stuck at, so it should not become the default intervention axis.

3.8 Predictive ability and risk segmentation

Predictive ability is read according to the incremental value of each layer of variables. The model using only TBA provides the baseline; when AE and BSE are added, Q^2_predict increases and RMSE decreases; when the interaction and control variables are added, RMSE continues to decrease. A positive Q^2_predict indicates that the model predicts better than the mean-value prediction benchmark, while RMSE allows comparison of errors among the three specifications being considered. Therefore, the result supports only the relative conclusion that the full specification has the lowest error among the three models; it does not turn PLSpredict into causal evidence.

Table 9: Comparison of three predictive specifications using PLSpredict

Model	Q^2 predict	RMSE	Relative assessment
M1: TBA	0,386	0,783	Baseline
M2: TBA + AE + BSE	0,599	0,633	Adds friction and capability
M3: + AE×BSE + controls	0,776	0,473	Lowest error

Source: PLSpredict results from the analytical dataset, n = 1.187, year 2026.

K-Means on the same 1.187 records produces three clusters with sizes of 504, 364, and 319 records, respectively accounting for 42,5%, 30,7%, and 26,9% of the sample. Clustering is retained only as an exploratory layer because Silhouette = 0,127 shows that the boundaries between clusters are not sharp. Therefore, the clusters are not treated as fixed customer segments or as a basis for assigning a risk label to an individual. The appropriate use is to view them as three temporary experience states for selecting pilot situations: a stable state, a low-trust–low-transparency state, and a high-friction–low-self-handling-capability state.

Table 10: Three experience states used for intervention segmentation

Cluster	n	Characteristics	Priority
Cluster 1	504	High TBA/CI; good BAR/BSE; low AE	Maintain stability
Cluster 2	364	Low TBA; low PT/BAR; high BPC	Transparency, privacy, and stability
Cluster 3	319	High AE; low BSE; low CI	Reduce retries, open contextual support

Source: Exploratory clustering results from the analytical dataset, n = 1.187, year 2026; results are used only as reference segmentation signals.

3.9 Robustness checks and synthesis of evidence

Robustness is checked using indicators that address different risks rather than stacking many techniques onto the same question. For the TBA equation alone, the inner VIF values of PSE, BAR, PT, and BPC are only 1,007 to 1,011; when the full CI model and control variables are considered, the largest VIF is approximately 1,93. These two levels indicate that the results are not dominated by severe multicollinearity. The maximum HTMT of 0,674 continues to preserve the boundaries between constructs. The block of control variables increases the R^2 of CI by about 0,060 but is not turned into a new hypothesis; the function of this block is to test whether the main relationships remain robust when age, device, number of re-authentication attempts, frequency, and transaction value are taken into account.

Table 11: Robustness checks and interpretive limits

Check layer	Result	Significance
Largest inner VIF	1,929	No substantial multicollinearity
Largest HTMT	0,674	Discriminant validity retained
ΔR^2 of control block	0,060	Background variables add information but do not replace their theoretical role
Power	0,958	Meets the design target of 0,95
Analytical design	Cross-sectional	Do not interpret relationships as causal

Source: Synthesis of robustness-check results from the analytical dataset, n = 1.187, year 2026.

The most important boundary does not lie in a statistical indicator but in the cross-sectional design. The coefficients indicate the degree of association in the research sample; they do not prove that changing an interface node will produce exactly the same increase in CI in actual operations. An A/B test by error code or a before–after design on the same users would come closer to the causal question.

3.10 Synthesis of research-model test results

The seven hypotheses form a consistent story rather than seven separate findings. The first four signals shape the extent to which customers can rely on the authentication step; TBA then accompanies continuance intention, while AE creates a separate cost. H7 places a boundary condition on that cost: the higher the self-efficacy, the less friction is translated into intention to leave. Therefore, the model does not say "security is the most important" or "ease of use is the most important"; it says that security, trust, and effort sit at different layers of the same decision.

Table 12: Summary of H1–H7 test results

Hypothesis	Relationship	β	Conclusion
H1	PSE → TBA	0,493	Positive direction; <0,001
H2	BAR → TBA	0,450	Positive direction; <0,001
H3	PT → TBA	0,428	Positive direction; <0,001
H4	BPC → TBA	-0,379	Negative direction; <0,001
H5	AE → CI	-0,363	Negative direction; <0,001
H6	TBA → CI	0,532	Positive direction; <0,001
H7	AE × BSE → CI	0,354	Positive direction; <0,001

Source: Summary of structural-model results from the analytical dataset, n = 1.187, year 2026.

4. CONCLUSION AND MANAGERIAL IMPLICATIONS

4.1 Conclusion

The chain of evidence answers four questions. First, trust in biometric authentication is not created by a single attribute. PSE → TBA reaches β=0,493; BAR → TBA β=0,450; PT → TBA β=0,428; BPC → TBA β=-0,379, and the R^2 of TBA reaches 0,769. All four β values are standardized coefficients; a sum of absolute values greater than 1 is not a sign of multicollinearity when the inner VIF values of these same four predictors are only around 1,01. Customers simultaneously evaluate whether the authentication step protects them, whether it operates stably, whether what is happening is explained, and whether sensitive data remain within a scope they feel they can control. PSE has the largest coefficient, but managerial priority still has to be considered together with the current state and the performance gap.

Second, trust does not erase operational cost. TBA → CI reaches β=0,532, while AE → CI reaches β=-0,363; the R^2 of CI reaches 0,780. A customer may trust that biometrics helps protect the account yet still reduce use if each transaction requires retries, long waiting, or self-guessing how to escape an error. The two results clearly separate "I trust the system" from "I am willing to continue expending effort to use the system." Therefore, increasing trust at the cognitive level cannot fully compensate for an authentication journey that continuously disrupts the transaction objective.

Third, the relationship between effort and continuance intention is not the same for all users. The AE × BSE interaction reaches β=0,354 and increases the R^2 of CI by about 0,124, while the component effect BSE → CI has f^2=0,006. When BSE is low, the AE → CI slope is about -0,717; at the medium level it is -0,363; when BSE is high, the slope is close to 0. Skilled users do not like a complex process more; they are simply less disrupted because they know how to adjust, retry, and follow instructions. The same story appears in the number of re-authentication attempts: CI decreases from 4,215 in the group that never had to repeat the process to 2,997 in the group that had to repeat it four or more times (η^2=0,359), while BAR decreases from 4,197 to

2,842 (η^2=0,393). Friction is therefore attached to an observable and fixable operational state, not merely to a general feeling.

Fourth, current perception levels show that the levers are not in the same state. After reverse-coding the two adverse constructs so that all higher scores carry a favorable meaning, BPC* reaches only 2,699 and AE* reaches 2,963; PT reaches 3,366, BAR 3,565, TBA 3,579, BSE 3,650, CI 3,681, while PSE has moved close to the agreement benchmark of 4 at 3,890. BPC therefore has the largest performance gap, but AE still ranks first on the priority index because it combines a large gap with a higher total effect on CI. The managerial order therefore cannot simply copy the β ranking, nor can it look only at the mean or gap in isolation.

The load-bearing point of the study lies at the boundary between necessary friction and friction that adds no security. A strict authentication step can still be accepted when customers understand what it protects, the system operates stably enough, and, if it fails, there is a clear path back. The cost becomes difficult to accept when a legitimate user must repeatedly provide the same evidence simply because the process has not handled its own uncertainty. In digital banking, customers do not need to close their accounts to react; they only need to gradually shift everyday transactions to a place that causes less disruption. Retention risk begins when the effort customers must pay no longer adds security.

The scope of the conclusion is kept consistent with the cross-sectional design. The coefficients describe relationships within the research sample; they do not prove that changing an interface node will create exactly the same increase in CI in real operations. The value of the results lies in identifying relationships that should be prioritized for further verification using operational data, telemetry, and controlled A/B testing.

4.2 Managerial implications in order of priority

Table 13: Total effect, performance, gap, and intervention priority

Variable	Total effect	Performance (%)	Gap (%)	Score	Rank
AE	-0,3645	49,1	50,9	0,1856	1
BPC	-0,2015	42,5	57,5	0,1159	2
PT	0,2276	59,1	40,9	0,0930	3
BAR	0,2390	64,1	35,9	0,0857	4
PSE	0,2620	72,3	27,7	0,0727	5

Source: Importance–performance analysis results from the analytical dataset, n = 1.187, year 2026.

Following the IPMA logic of Ringle and Sarstedt (2016), managerial priority should be read simultaneously from the level of importance to the target variable and current performance. Table 13 uses the total effect on CI as the measure of importance, converts performance to a 0-100 scale, and calculates Gap = 100 - Performance. To obtain a single implementation order, the priority score is calculated as |total effect| × Gap/100; this is a composite index used to rank interventions, not a mandatory standard IPMA indicator. AE ranks first because it has both high importance and a substantial remaining gap; BPC ranks second despite having the largest gap because its total effect on CI is lower. PT, BAR, and PSE follow in that order. This ranking is a starting point for allocating attention, not a rigid technical queue: a lower-ranked cause can still be corrected in the same wave if it directly generates the higher-priority pain point.

Wave 1 - Reduce friction and fix where friction originates: AE + BAR

The first wave focuses on Perceived Authentication Effort (AE), but does not address AE by mechanically cutting screens. Customers expend the most effort when they have to rescan, lose a previously valid state, wait and then return to an earlier step, or do not know why the previous attempt failed. MB needs to record the authentication flow event by event and standardize a minimum error-code taxonomy for facial recognition, NFC/identity-card reading, connection, timeout, App version, and state loss. When part of the data is already valid, the application should preserve that part; when the same error repeats, the system needs to switch to suitable guidance or an alternative handling path instead of forcing the customer to start the entire process again.

BAR must be addressed in the same Wave 1 even though its priority rank is lower than AE, because technical stability may itself be the source that creates loops and raises AE. The load-bearing KPIs for this wave are first-pass success, median number of attempts, P90 completion time, the proportion of sessions requiring three or more attempts, abandonment rate, and state-loss rate; the indicators are separated by error type, device, and operating-system version. An intervention is expanded only when friction decreases without worsening safety guardrails such as false acceptance, risky transactions, and unauthorized-access incidents.

Wave 2 - Reduce information uncertainty and restore a sense of control: BPC + PT

After the authentication flow becomes more stable, the remaining cost is not only "I keep trying and still cannot finish" but also "what data are being used, why do I have to do this step, and where do I go next if it fails?" BPC ranks second in priority and PT ranks third, so the two pillars are addressed together in Wave 2. For BPC, the App needs to place a short information layer at the point where biometrics are requested, stating the data being used, the purpose of use, and the channel for checking or requesting support; lengthy legal content is pushed down to the detail layer. For PT, error messages need three short parts: the cause identified by the system, the next action, and an exit path when retrying is no longer reasonable.

The KPIs for Wave 2 must measure whether customers have to guess less: self-recovery rate after an error, success rate on the next attempt, time to exit the error state, rate of transfer to a support channel, and PT, BPC, and TBA levels after the information is clarified. The objective is not to make customers read more, but to reduce the number of times they must infer for themselves what the system is doing and where their data are.

Maintain - Keep PSE high without buying an additional sense of security through friction

Perceived Security Effectiveness (PSE) is placed in the Maintain layer not because it is less important, but because Table 13 shows that PSE has less room for improvement than the remaining pain points. MB should maintain the sense of protection through short, timely messages tied to the specific risk that the authentication step is preventing; it should not add operations merely to make the process "look more secure." The KPIs for the Maintain layer are that PSE and TBA do not decline while AE, completion time, and retry rate continue to decrease. Any change that makes the experience faster but worsens safety indicators is not considered an improvement.

BSE - A continuous support layer, not a separate Wave

Biometric Self-Efficacy (BSE) plays a moderating role and therefore does not compete in ranking with the other five intervention pillars. The results show that the direct effect of BSE on CI is very small, but the AE × BSE interaction is clear; BSE is therefore used to decide when the system needs to provide more support: remaining too long at one step, retrying many times, opening help, or repeating the same type of error. KPIs should track the gaps in first-pass success, retries, completion time, and channel-switching rate between low-and high-BSE groups; only a narrowing gap shows that the experience is becoming less dependent on customers' ability to fend for themselves.

90-day roadmap and verification principles

The first thirty days are used to lock the error-code taxonomy and establish baselines for first-pass success, retries, P90 time, abandonment, PT, BPC, and safety guardrails. Days 31-60 run controlled pilots for four intervention groups: preserving state after an error and an alternative handling path in Wave 1; diagnostic messages and a privacy-information layer in Wave 2; adaptive support according to BSE; and PSE messages placed only at authentication points that have become stable. Days 61-90 expand only interventions that produce repeated improvement over at least two measurement cycles without worsening guardrails. The KPI chain follows three layers—operations, experience, and outcomes—to distinguish a change that genuinely helps customers complete the process more easily from a change that merely improves the appearance of a technical indicator.

4.3 Organization of verification and implementation-risk control

The clustering in Table 10 uses only one K-Means solution on 1.187 records: Cluster 1 contains 504 records, Cluster 2 contains 364 records, and Cluster 3 contains 319 records. These three figures are used consistently throughout the results and recommendations. However, Silhouette reaches only 0,127, so the boundaries between clusters still overlap; the clusters are therefore used only to select verification situations, not to assign fixed labels to customers. A person can change state after one error, an App update, or a transaction of a different value; what needs to be monitored is the state they are encountering in the authentication session, not a "risk profile" permanently attached to that person.

The organization of interventions must therefore move from event to cause. When first-pass success decreases, the system needs to trace whether the error comes from NFC, camera, network, App version, or business logic; when TBA decreases but technical errors do not increase, the focus shifts to PT, BPC, and information at the authentication point. If MB looks only at the final transaction success rate, it will overlook the cost customers have already paid before reaching that outcome.

Each change should have three control layers. The safety layer monitors fraud, correctly blocked risky transactions, and unauthorized-access incidents; the experience layer monitors AE, BAR, number of attempts, and completion time; the behavioral layer monitors TBA, CI, and the rate of transactions continuing on the App. A change is expanded only when it reduces friction without weakening the safety layer, because a faster process that lets more risky transactions slip through is not an improvement.

Table 14: Operational indicator set linked to priorities

Pillar	Monitoring KPI	Direction
AE	First-attempt success; number of attempts; P90 time; abandonment	↑; ↓; ↓; ↓
BPC	Rate of viewing/understanding data information; privacy complaints	↑; ↓
PT	Rate of errors with actionable guidance; post-error recovery rate	↑; ↑
BAR	Error rate by device/NFC/camera; state loss	↓; ↓
PSE	Rate of risky transactions correctly blocked; TBA after warning	Maintain; ↑
BSE	Self-recovery rate in the low-BSE group; transfer to support channel	↑; ↓

Source: Synthesis of research results from the analytical dataset, n = 1.187, year 2026.

4.4 Decision-making principles and guardrails

The KPI system follows the operations–experience–outcomes chain. The operational layer tracks first-pass success, number of attempts, P90 time, errors by device, and alternative-path switching rate; the experience layer tracks AE, BAR, PT, BPC, and TBA; the outcome layer tracks CI together with transaction behavior on the App. The three layers allow the break point to be traced backward: if technical errors are stable but TBA decreases, transparency or privacy needs to be examined; if TBA increases but CI does not, the remaining part may lie in App quality, products, fees, or switching costs outside the authentication layer.

An intervention is expanded only when it simultaneously satisfies three conditions: the target indicator improves, safety guardrails do not worsen, and the effect is repeated across at least two measurement cycles. If AE decreases but risky transactions increase, that improvement is rejected; if first-pass success increases but false acceptance increases, the system is trading security for convenience; if TBA increases but CI and transaction behavior do not move, the authentication layer may have exhausted its ability to explain the remainder. Predefined stopping rules help MB avoid continuing to pour resources into an indicator that looks better but no longer pulls the final outcome.

4.5 Expansion criteria after each verification cycle

At the end of each implementation cycle, MB needs to answer six questions before expanding: which source creates the most AE; which intervention reduces P90 and increases first-pass success; whether changes in PT and BPC accompany recovery of TBA after an incident; whether the low-BSE group reduces its need to switch to support; whether CI in a short survey moves; and whether post-intervention transaction behavior changes. The six questions force decisions to pass through all three layers of operations, experience, and behavior, instead of declaring success merely because one technical indicator increases.

The overall average must not be used to conceal long-tail error groups. First-pass success may be high while one device group still has to retry many times; P90 completion time and the proportion of sessions with three or more attempts therefore need to be read alongside the overall success rate. The same principle applies to privacy: a decline in the average BPC score is meaningful only when correct understanding of the collection purpose increases, the support channel becomes easier to find, and complaints are closed within the correct SLA. If the concern score falls because users receive less information, the indicator looks better while the actual quality of control worsens.

The order of priority should also not be fixed after a single analysis. If AE decreases sharply but BAR remains low, the next focus must shift to stability; if BAR increases but TBA does not recover, PT or BPC is the new bottleneck; if TBA increases but CI remains unchanged, the authentication layer has exhausted its ability to explain the remainder and MB must examine application quality, products, fees, or alternatives.

APPENDIX 1: QUANTITATIVE RESEARCH QUESTIONNAIRE
SURVEY ON THE BIOMETRIC AUTHENTICATION EXPERIENCE ON THE MBBANK APP

Dear Sir/Madam,

I am currently conducting the research topic ***"Research on factors influencing Trust in biometric authentication and Continuance intention to use the MBBank App among individual customers in Ho Chi Min****h City." I would greatly appreciate your support by sharing your actual experience through this survey questionnaire.*

Please answer the questions according to your own experience; there are no right or wrong answers. The survey is intended for customers aged 18 or older, currently living or working in Ho Chi Minh City, who have used the MBBank App and completed biometric authentication at least 3 times in the past 6 months. The survey does not request a facial image, identity-card number, account number, card number, password, or OTP code. The collected data are used solely for research purposes, are aggregated anonymously, and are not used for commercial purposes. You may stop answering at any time. Sincerely thank you for your cooperation!

PART I. Please provide the following information:

To ensure the suitability of the survey, please answer several screening questions below.

Screening question 1: Are you at least 18 years old?

- ☐ Yes *(Please continue answering the following questions)*
- ☐ No (W*e are sorry, this survey is not suitable for you. Thank you for your interest!)*

Screening question 2: Are you currently living or working mainly in Ho Chi Minh City?

- ☐ Yes *(Please continue answering the following questions)*
- ☐ No (W*e are sorry, this survey is not suitable for you. Thank you for your interest!)*

Screening question 3: Are you currently using the MBBank App?

- ☐ Yes (Please continue answering the following questions)
- ☐ No (We are sorry, this survey is not suitable for you. Thank you for your interest!)

Screening question 4: In the past 6 months, have you scanned your face or a chip-based identity card on the MBBank App at least 3 times?

- ☐ Yes *(Please continue answering the following questions)*
- ☐ No (W*e are sorry, this survey is not suitable for you. Thank you for your interest!)*

1. **How long have you used the MBBank App? (Select 1)**

   ☐ 1. Under 6 months
   ☐ 2. From 6 months to under 1 year
   ☐ 3. From 1 year to under 3 years
   ☐ 4. From 3 years to under 5 years
   ☐ 5. From 5 years or more

2. **On average, how frequently do you transact on the MBBank App? (Select 1)**

   ☐ 1. Less than 1 time/week
   ☐ 2. 1–2 times/week
   ☐ 3. 3–5 times/week

☐ 4. 6 times or more/week

3. What is the value of a transaction that you typically make on the MBBank App? (Select 1)

☐ 1. Under 1 million VND
☐ 2. From 1 to under 10 million VND
☐ 3. From 10 to under 20 million VND
☐ 4. From 20 million VND or more

4. When placing the chip-based identity card close to the back of the phone as instructed by the MBBank App, can your phone read the information? (Select 1)

☐ 1. Yes
☐ 2. No
☐ 3. Have never performed it or do not remember

5. In the past 6 months, how many times have you had to repeat the process because the App failed to authenticate successfully? (Select 1)

☐ 1. None
☐ 2. 1 time
☐ 3. 2–3 times
☐ 4. 4 times or more

PART II. Please indicate your perceptions of the biometric authentication experience on the MBBank App according to the statements below.

I would like to hear your perceptions based on your experience using the MBBank App during the most recent six months. In the statements below, the "authentication step" means the facial scan or chip-based identity-card reading step required by the App.

For each statement, please mark X in one box fr**om** 1 to 5. By conventio**n**, a higher number indicates a higher level of agreement, specifically as follows:

Strongly dis agree	Disa gree	Neither agree nor disagree	Agree	Strongly agree
1	2	3	4	5

No.	Statements	Level of agreement				
		Strongl y disagree	Disa gree	Neither agree nor disagree	Agree	Strongly agree
1. Perceived Security Effectiveness (PSE)						
PSE1	The biometric authentication step helps prevent other people from logging into my MBBank account.					
PSE2	This authentication step helps reduce the risk of other people impersonating me to make transactions.					
PSE3	This authentication step helps prevent other people from making transactions without my permission.					
PSE4	If another person gains possession of my phone, this authentication step still helps protect the account.					
2. Biometric Authentication Reliability (BAR)						
BAR1	The MBBank App correctly recognizes my face.					
BAR2	The facial scanning step on the MBBank App operates stably.					

BAR3	I usually authenticate successfully on the first attempt.					
BAR4	The authentication process is usually completed without being interrupted midway.					
3. Authentication Process Transparency (PT)						
PT1	MBBank clearly tells me why I have to complete biometric authentication.					
PT2	MBBank tells me which data are used during authentication.					
PT3	MBBank tells me the purposes for which authentication data are used.					
PT4	When authentication is unsuccessful, MBBank clearly guides me on what I need to do next.					
4. Biometric Privacy Concern (BPC)						
BPC1	I am concerned that my facial data may be accessed without authorization.					
BPC2	I am concerned that the data used for my authentication may be used for the wrong purpose.					
BPC3	I am concerned that the data used for my authentication may be shared without my consent.					
BPC4	I am concerned that if the data used for my authentication are leaked, the consequences may be long-lasting.					
5. Perceived Authentication Effort (AE)						
AE1	I have to perform many steps to complete authentication.					
AE2	I spend a lot of time completing one authentication.					
AE3	I have to concentrate a lot to perform the authentication steps correctly.					
AE4	I find authentication requires a lot of effort.					
6. Biometric Self-Efficacy (BSE)						
BSE1	I can complete the authentication steps on the MBBank App by myself.					
BSE2	If the first authentication attempt is unsuccessful, I know how to try again.					
BSE3	I know how to position my face correctly according to the instructions on the screen.					
BSE4	I can follow the on-screen instructions by myself when authentication is unsuccessful.					
7. Trust in Biometric Authentication (TBA)						
TBA1	I trust the biometric authentication step on the MBBank App.					
TBA2	I trust that this authentication step works as MBBank states.					
TBA3	I feel reassured when a transaction includes a biometric authentication step.					
TBA4	I trust that MBBank carries out biometric authentication responsibly.					
8. Continuance Intention to Use the MBBank App (CI)						
CI1	I will continue using the MBBank App during the next six months.					
CI2	I will continue using the MBBank App to conduct banking transactions.					
CI3	When a transaction requires biometric authentication, I will still conduct it on the MBBank App.					
CI4	The MBBank App will remain a banking application that I use regularly.					

PART III. Please provide some general information about yourself:

1. Gender:

☐ 1. Male☐ 2. Female

2. Your age:

☐ 18–25 years old ☐ 26–35 years old

☐ 36–45 years old ☐ 46–55 years old

☐ From 56 years old or above

PART IV. Open-ended questions (Optional):

1. Among the authentication steps on the MBBank App, which step do you find the most difficult? Please describe briefly.

....

....

2. What should MBBank change first to make authentication easier?

....

....

Please do not provide your account number, password, OTP code, or identity-card data in the open-ended response section.

Sincerely thank you for your cooperation!

APPENDIX 2: RESEARCH RESULTS

Verification results file: MB Bank results.xlxs

Verification technique (Python): MBBank_Research_Analysis.py